

or unlock the gate fitted with an electric door locking system.

The complete receiver circuit is shown in Fig. 2.

Power supply. The circuit requires 12V, 5V and 3.3V DC supplies for various sections. 12V supply is derived from a 230V AC primary to 15V, 500mA secondary transformer. The output from bridge rectifier module BR1 is filtered by capacitor C1 and then fed to 7812 regulator. 12V output from 7812 regulator IC is given to 7805 regulator IC and electric lock and its driver section. 5V from the output of 7805 IC is given to the AVR microcontroller. Zener diode ZD1 (3.3 V) is used to obtain 3.3V supply for the Bluetooth module.

The glowing of LED1 indicates the presence of power in the circuit.

AVR microcontroller. AVR microcontroller is the heart of the system. The program in the microcontroller processes the received signal and then locks or unlocks the electric lock.

The Atmel ATmega328 is a low-power CMOS 8-bit microcontroller based on the AVR enhanced RISC architecture. By combining an 8-bit RISC CPU with in-system self-programmable Flash on a monolithic chip, the ATmega328 is a powerful microcontroller that provides a highly flexible and cost-effective solution to many embedded control applications. It has a rich instruction set with 32 general-purpose working registers.

Some of its features are 32 kB of in-system programmable Flash with read-while-write capabilities, 1kB EEPROM, 2kB SRAM, 23 general-purpose input/output (I/O) lines, three flexible timers/counters with compare modes, internal and external interrupts, a serial programmable USART, a byte-oriented two-wire serial interface, an SPI serial port, a 6-/8-channel 10-bit analogue-to-digital converter (ADC), a programmable watchdog timer with internal oscillator, and five software-selectable power saving modes.

The on-chip ISP Flash allows the program memory to be reprogrammed in-system through an SPI serial interface, by a conventional non-volatile



Fig. 3: HC-05 Bluetooth module



Fig. 4: Electric strike YS 130



Fig. 5: An electronic door locking system (Courtesy: elektrokrats.com)

memory programmer or an on-chip boot program running on the AVR core. The boot program can use any interface to download the application program in the application Flash memory. Software in the boot Flash section continues to run while the application Flash section is updated, providing true read-while-write operation.

The AVR communicates with the Bluetooth module through its TXD (transmit) and RXD (receive) pins. The signal received from the Bluetooth module is processed by the software in the microcontroller. The control output signal is available at its pin 15.

Bluetooth module. We have used a 34-pin, HC-05 Bluetooth module in this project. It is an easy-to-use Bluetooth

serial port protocol (SPP) module, designed for transparent wireless serial connection setup.

Serial port Bluetooth module is fully qualified for Bluetooth V2.0+EDR (enhanced data rate) 3Mbps modulation with complete 2.4GHz radio transceiver and baseband. It is a low-power module that operates off 1.8 V to 3.6 V. Its default values are: baud rate-38,400, data bits-8, stop bit-1, parity-no. It supports baud rates of 9600, 19,200, 38,400, 57,600, 115,200, 230,400 and 460,800. By default

the auto-pairing pin code is set as '0000' and password as 1234. The HC-05 Bluetooth module is shown in Fig. 3.

Care should be taken while handling this module. You need to extend four connections to the circuit. These connections are wires for TXD, RXD, V_{CC} and GND pins. TXD pin of the module should be connected to RXD pin of the microcontroller, and RXD pin should be connected to TXD pin of the microcontroller. V_{CC} pin is connected to 3.3 V and GND pin is connected to the common ground rail of the circuit.

Electric strike. Electric strike (refer Fig. 4) is an access control device used to lock and release doors. Its applications can be found in all areas ranging from single door to large PC-based systems.

Electric strikes are installed in or on the door frame and work in conjunction with mechanical door locks. All strikes basically work on the principle of electronically controlling the temporary free movement of the jaw (striker) allowing door opening without manual retraction of the latch-bolt. A typical electronic door locking system is shown in Fig. 5.

Electric strikes are generally available in two configurations:

1. Fail-secure. In this configuration, applying electric current to the strike